

Notes: Young (QJE, 2000)

The Razor's Edge: Distortions and Incremental Reform in the People's Republic of China

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1 Big picture

- **Question.** Can gradual or incremental reform in a centrally planned economy generate *new distortions* rather than simply removing old ones? In particular, did China's post-1978 reforms create incentives for local governments to fragment the domestic market and distort regional production?
- **Setting.** The paper studies reform-era China, with a special focus on **interregional industrial duplication, price dispersion, and factor allocations across provinces** from the Maoist period through the reform era.
- **Core challenge.** Anecdotes about local protectionism and trade wars are not enough. The paper needs to show, using aggregate data, that the Chinese economy became more internally fragmented in ways that are hard to reconcile with standard stories of market liberalization, openness, or comparative advantage.
- **Main argument.** Under the plan, prices were distorted so that profits and fiscal revenue were concentrated in industry. As central controls over factor allocation weakened, local governments sought to capture those industrial rents by building up local processing and manufacturing capacity. Once these local industrial interests existed, governments had incentives to protect them through administrative and physical barriers to trade.
- **Main conceptual contribution.** The paper adds a **third pitfall of incremental reform** to the transition literature. Beyond standard second-best losses from partial reform and political backlash, gradual reform can trigger **rent-seeking responses** by partially liberalized actors, who then create new distortions to defend newly acquired rents.
- **Why it matters.** The paper argues that reform should not be interpreted simply as a one-way move from plan to market. In China, some elements of central planning were removed, but local governments replaced them with their own interventionist structures. That means rapid growth does not by itself prove that markets became more integrated or allocations more efficient.
- **Main empirical question.** Did China's reform era lead provinces to become **more similar in output structure, more different in prices and factor allocations**, and ultimately **more distorted relative to comparative advantage**?
- **Main empirical results.**
 - Provincial output structures converged strongly during the reform era, meaning provinces increasingly duplicated one another's industries.
 - Regional price dispersion rose sharply in the late 1980s and then fluctuated without clear reversal in the 1990s.
 - A variance decomposition shows that while relative outputs converged, relative labor productivities and labor allocations diverged.
 - In agriculture, the historical positive relation between labor intensity and grain yields disappeared during reform.
 - Provinces with better agricultural weather increasingly *withdrew* labor, fertilizer, irrigation, and machinery from agriculture, which is consistent with factor allocations moving *against* comparative advantage.

- **Economic takeaway.** Reform in China is best understood not as simple liberalization, but as a **devolution of planning power** from the center to local governments. The result was a more open international economy combined with a more fragmented internal economy.

2 Data and measurement (key objects)

2.1 Main data sources

- **Public finance and revenue data.** The paper uses Chinese budgetary statistics to study how government revenue was distributed across sectors and across central versus local levels.
- **Provincial output data.** Provincial GDP and earlier “national income” data by sector are taken from provincial statistical yearbooks, historical compilations, and the *China Statistical Yearbook*.
- **Regional price data.** The paper assembles four separate datasets on regional prices:
 - annual retail prices for **305 consumer goods** in **30 cities**, 1986–1993;
 - annual purchase prices for **130 agricultural goods** in **30 provinces**, 1986–1993;
 - monthly market prices for **49 industrial materials** in **36 cities**, 3/1990–5/1999;
 - monthly market prices for **33 agricultural products** in **36 cities**, 6/1993–5/1999.
- **Agricultural data.** The final part of the paper uses provincial grain output, sown area, irrigated area, fertilizer use, agricultural machinery power, and primary-sector employment.
- **Weather data.** Provincial rainfall and its variability are used to construct a proxy for weather-based comparative advantage in agriculture.

2.2 Regional units, sectors, and market objects

- **Regional unit.** The core unit is the **province** (or province-level administrative division).
- **Provincial coverage.**
 - The reform-period GDP analysis uses the **30 provincial units** in place in 1996.
 - The earlier “national income” series uses **28 provinces**, because Tibet and Hainan do not have full historical coverage in the long series.
 - The weather regressions use **29 provinces** beginning in 1985.
- **Sector definitions.**
 - In the older national-income data, output is split into **agriculture, industry, construction, transport, and commerce**.
 - In the GDP data, output is split into **primary, secondary, and tertiary** sectors.
 - In the variance decomposition, the focus is on **primary versus secondary industry**, because these are the sectors most naturally interpreted as tradable.
- **Price-market object.** For each product-time observation, the paper computes the cross-region dispersion of log prices.

- **Agricultural productivity object.** The paper measures grain yield as grain output per hectare sown, and compares this to factor intensity, especially labor per hectare and other input intensities.

2.3 Key descriptive facts

- **Industry dominated fiscal revenue under the plan.** Industry accounted for roughly **36%** of budgetary revenue in 1952, **73%** in 1965, and **75%** in 1978. Even in 1995, after reform, industry still accounted for about **44%** of budgetary revenue.
- **Revenue collection was historically local.** Before reform, so-called local revenue was many times larger than central revenue. Around the mid-1970s, local revenue was nearly **seven times** central revenue, even though most “central” revenue was in practice collected locally and then remitted upward.
- **Local governments had a strong fiscal interest in enterprise performance.** They depended on industrial revenue and were also responsible for many enterprise-linked social functions such as housing and welfare.
- **Rural industrialization was aggressively promoted.** New rural enterprises often received tax holidays, subsidized credit, and official encouragement, helping them capture rents created by plan-era price wedges.
- **Price data are noisy but informative.** The paper repeatedly emphasizes that Chinese regional price data are poor, incomplete, and unbalanced. The empirical argument therefore rests on broad patterns, not on overly precise price levels.

2.4 Measurement caveats and timing

- **Extra-budgetary finance is difficult to measure.** The paper excludes extra-budgetary funds from the main fiscal analysis because definitions changed over time and historical sectoral breakdowns are unavailable.
- **Output is measured in nominal terms.** The paper does not construct regional PPP deflators, so the level analysis of GDP by sector is not a real-output exercise in the modern spatial-price sense.
- **Price panels are unbalanced.** Products and reporting cities enter and leave the sample, which is why the paper uses time-dummy methods that exploit repeated observations of product-specific cross-regional dispersion.
- **Agricultural weather analysis starts only in 1985.** This matters because some of the deterioration in the relationship between yields and factor inputs had already begun before the regression sample starts.
- **Empirical timing.** The paper compares three broad periods:
 - the early postwar and pre-reform era,
 - the Cultural Revolution period, when industrial duplication accelerated,
 - the post-1978 reform era, especially the 1980s and 1990s.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual mechanism: from plan distortions to local protectionism

- The paper is not a structural model of firm behavior. Its “model” is a political-economy mechanism:
 1. plan-era **price scissors** make industry fiscally attractive;
 2. reforms devolve control over enterprises and budgets to local governments;
 3. local governments expand downstream industry to capture these rents;
 4. once those interests are created, local authorities use trade barriers and controls to protect them.
- **Interpretation.** The central mechanism is dynamic and path dependent. Distortions today alter who gains from reform tomorrow; those winners then lobby for or directly impose new distortions that protect their rents.

3.2 Convergence in the structure of output

Let s_{ijt} denote the share of sector j in province i 's output at time t , and let $\bar{s}_{jt}$ denote the average sector share across provinces. The paper studies convergence using the aggregate absolute and squared deviations

$$D_t^{\text{abs}} = \sum_i \sum_j |s_{ijt} - \bar{s}_{jt}|, \quad D_t^{\text{sq}} = \sum_i \sum_j (s_{ijt} - \bar{s}_{jt})^2.$$

It also considers weighted versions that weight each province by its share of total GDP.

Interpretation.

- If provinces specialize according to different comparative advantages, output shares should tend to *diverge*, or at least not systematically converge, unless comparative advantage itself converges.
- If output structures converge sharply, one interpretation is that provinces are duplicating one another's industries rather than specializing.
- The paper's main finding is that these dispersion measures fall markedly both during the Cultural Revolution and during the reform period.

3.3 Price-dispersion time-dummy regression

For each product p in each time period t , the paper computes the cross-region standard deviation of log prices, σ_{pt} . It then estimates a reduced-form specification of the form

$$\ln \sigma_{pt} = \alpha_p + \delta_t + \varepsilon_{pt},$$

where α_p is a product fixed effect and δ_t is a time dummy.

Interpretation.

- The product fixed effect absorbs permanent differences in how much price dispersion some goods naturally have.
- The time dummies trace the evolution of overall interregional price dispersion over time.
- Rising δ_t means that the same kinds of goods exhibit more cross-region price dispersion, which is consistent with more segmented markets.

3.4 Variance decomposition of relative output

Define relative output between primary and secondary sectors in province i as

$$R_i = \ln \left(\frac{P_{pi} Q_{pi}}{P_{si} Q_{si}} \right).$$

This can be decomposed into a relative nominal labor-productivity term and a relative labor-allocation term:

$$R_i = \ln \left(\frac{P_{pi} Q_{pi} / L_{pi}}{P_{si} Q_{si} / L_{si}} \right) + \ln \left(\frac{L_{pi}}{L_{si}} \right) \equiv A_i + B_i.$$

Across provinces,

$$\text{Var}(R_i) = \text{Var}(A_i) + \text{Var}(B_i) + 2 \text{Cov}(A_i, B_i).$$

Interpretation.

- A_i measures relative **nominal labor productivity** between primary and secondary sectors.
- B_i measures relative **labor allocation** between those sectors.
- If convergence in output shares were driven by market integration or converging comparative advantage, one would expect labor allocations and productivities to become *more similar* across provinces.
- The paper instead finds that output convergence occurs while the dispersion in A_i and B_i rises, offset by a more negative covariance. That pattern is hard to reconcile with clean market-based specialization.

3.5 Agriculture: factor intensity and grain yields

The paper first studies the raw relation between grain yields and labor intensity across provinces:

$$\text{Yield}_{it} = \frac{\text{grain output}_{it}}{\text{grain sown area}_{it}}, \quad \text{Labor intensity}_{it} = \frac{\text{primary employment}_{it}}{\text{sown area}_{it}}.$$

Interpretation.

- In a standard productivity story, higher labor intensity should be associated with higher yields, at least weakly.
- Historically this positive relation was present in Chinese data.
- Its disappearance during the reform era is interpreted as evidence that factor allocations were no longer moving toward productive agricultural regions.

3.6 Weather-based comparative-advantage regressions

The core yield regression is

$$\begin{aligned}\ln(\text{Yield}_{it}) = & \beta_0 + \beta_1 \ln(\text{Irri}_{it}) + \beta_2 \ln(\text{Fert}_{it}) + \beta_3 \ln(\text{Power}_{it}) \\ & + \beta_4 \ln(\text{Labor}_{it}) + \beta_5 \text{Prec}_{it} + \beta_6 \text{Prec}_{it}^2 + \beta_7 \text{CVPrec}_{it} + \alpha_i + \gamma t + \varepsilon_{it},\end{aligned}$$

where irrigation, fertilizer, machinery power, and labor are normalized by sown area.

From the rainfall terms, the paper constructs a synthetic provincial weather measure

$$\text{Weather}_i = \frac{\hat{\beta}_5 \text{Prec}_{it} + \hat{\beta}_6 \text{Prec}_{it}^2 + \hat{\beta}_7 \text{CVPrec}_{it}}{\text{...}}$$

and then studies whether provinces with better weather move resources into or out of agriculture over time using

$$X_{it} = a + bt + c \text{Weather}_i + d(\text{Weather}_i \times (t - 1985)) + \alpha_i + u_{it},$$

where X_{it} is alternatively irrigation, fertilizer, power, labor, agriculture's labor share, grain output per capita, or yields.

Interpretation.

- The rainfall terms identify a time-invariant proxy for agricultural comparative advantage.
- The interaction coefficient d asks whether provinces with more favorable weather increasingly shift factors *toward* or *away from* agriculture over time.
- The key empirical result is $d < 0$ for the major agricultural inputs, which means that provinces with better weather increasingly withdrew factors from agriculture.

4 Identification and instruments (what makes the estimates credible)

4.1 What simple anecdotes cannot establish

- Stories about “trade wars” and local protectionism are suggestive, but they are not sufficient to show that the domestic market became more fragmented.
- Anecdotes can always be countered with opposite anecdotes about interregional cooperation or efficiency-enhancing local intervention.
- The paper therefore treats the anecdotal evidence as motivation only, not as identification.

4.2 Why output convergence is suggestive but not decisive

- If provinces increasingly resemble one another in sectoral output shares, that is consistent with industrial duplication and local attempts to build self-sufficiency.
- But output convergence alone does not prove fragmentation. It could also reflect:
 - opening to the world economy, if Chinese provinces share comparative advantage relative to the rest of the world;

- convergence in factor endowments or technologies across provinces;
- changes in sector classifications or measurement.
- This is why the paper layers additional evidence on top of output convergence.

4.3 Why price dispersion matters

- If barriers to trade segment regional markets, then the same goods should display more interregional price dispersion.
- That said, price divergence is also not decisive on its own. It could reflect poor data quality, product-quality differentiation, or temporary local shocks.
- The credibility of the price argument comes from its timing and its consistency with the institutional story: dispersion rose in the late 1980s and failed to clearly reverse in the 1990s, despite continuing reform rhetoric.

4.4 What the variance decomposition rules out

- The variance decomposition is the paper’s most important rebuttal to the “maybe this is just comparative advantage” critique.
- Under a conventional specialization story, convergence in relative outputs should be accompanied by convergence in either labor allocations, labor productivities, or both.
- China exhibits the opposite pattern during reform: output structure converges while labor allocations and productivities diverge.
- The U.S. benchmark and the earlier Chinese postwar benchmark strengthen the argument. In those comparison cases, output convergence came with declining dispersion in labor allocations, which is what a standard market story would predict.

4.5 Why the weather design is the strongest evidence – and its limits

- Weather is valuable because it is plausibly exogenous to reform-era local industrial policy and, at least over the relevant horizon, captures a persistent source of agricultural comparative advantage.
- If regions with better agricultural weather increasingly remove labor and complementary inputs from agriculture, that is direct evidence of factor allocation moving against comparative advantage.
- But the paper is candid about limitations:
 - weather is only an imperfect proxy for comparative advantage;
 - omitted inputs can bias the yield regressions;
 - the author cannot fully solve endogeneity problems with the available data.
- So the weather exercise is best read as **supportive and triangulating evidence**, not as a clean quasi-experiment.

4.6 Bottom line on empirical credibility

- The paper's credibility comes from **triangulation across many margins**:
 1. the fiscal and institutional story in section II,
 2. convergence in output composition,
 3. divergence or non-convergence in price dispersion,
 4. worsening variance-covariance patterns in sectoral productivity and labor allocation,
 5. and agriculture-weather evidence showing factor movements against comparative advantage.
- None of these pieces alone is conclusive, but together they all point in the same direction: internal fragmentation and distortionary local planning increased during reform.

5 Tables and figures

5.1 Table I and Figure I: industry as the fiscal core and the local basis of control

TABLE I
DISTRIBUTION OF GOVERNMENT BUDGETARY REVENUE BY SECTOR (SHARES)

	Industry	Agriculture	Commerce	Transportation	Construction	Other
1952	0.36	0.21	0.24	0.05	0.00	0.13
1965	0.73	0.07	0.07	0.09	0.00	0.04
1978	0.75	0.03	0.12	0.07	0.00	0.03
1995	0.44	0.06	0.33	0.04	0.02	0.12

Industry includes mining, manufacturing, and utilities. Revenue is classified according to the nature of the institution, not the type of revenue; e.g., "industry" includes nonindustrial revenue collected by industrial ministries.

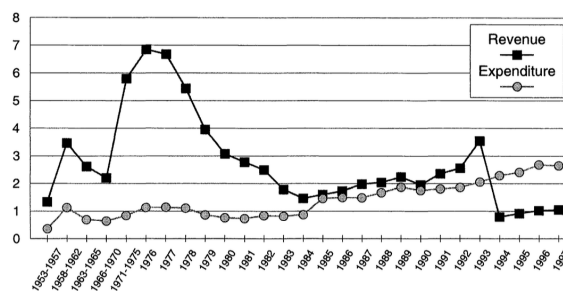
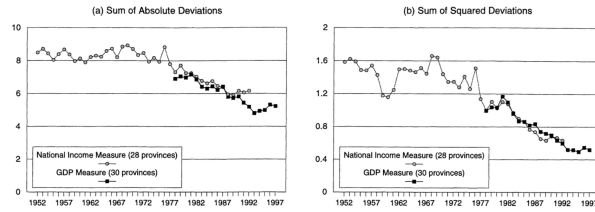


FIGURE I
Ratio of Local to Central Revenue and Expenditure

- **Read.** Table I shows that industry was the principal source of budgetary revenue under the plan: its revenue share rose from **0.36** in 1952 to **0.73** in 1965 and **0.75** in 1978, before falling to **0.44** by 1995.
- Figure I shows that revenue collection was overwhelmingly local in practice, with the ratio of local to central revenue very high before reform.
- **Main point.** Local governments had both the information and the fiscal incentives to care deeply about industrial activity. This makes the later emergence of local industrial protectionism politically plausible.

5.2 Figure II and Table II: convergence in provincial output structure

- **Read.** Figure II plots the sum of absolute and squared deviations of provincial output shares from the average structure. Both the older national-income series and the reform-era GDP series show strong convergence.



Note: National Income divided into five sectors: agriculture, industry (mining, manufacturing and utilities), construction, transport and commerce (wholesale and retail trade and catering). GDP divided into three sectors: primary (agriculture), secondary (mining, manufacturing, construction and utilities) and tertiary (services).

FIGURE II
Convergence in the Composition of Output (deviation from average production structure in current prices)

- Between **1969 and 1978**, the sum of absolute deviations in the national-income series falls by about **18%**. During reform it falls further. In the GDP series, the sum of absolute deviations falls by about **25%** by 1992.
- The squared-deviation measures show even stronger convergence: the national-income measure falls by about **39%** between 1969 and 1978, and the GDP measure falls a further **48%** by 1997.
- Table II shows that these results are robust to using weighted deviations, deviations from the national rather than provincial average structure, and separate coastal/interior decompositions.
- **Main point.** China's provinces increasingly duplicated one another's sectoral composition during reform.

5.3 Figure III and Figure IV: sector shares versus absolute output levels

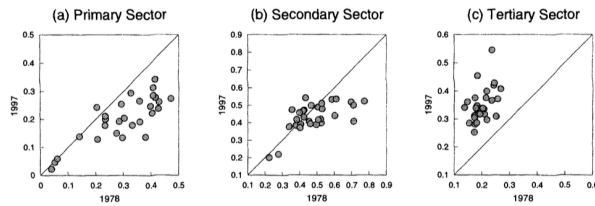


FIGURE III
Shares of GDP (30 provinces)

TABLE II
CONVERGENCE IN THE STRUCTURE OF OUTPUT

Deviations from provincial average structure:

	Absolute						Squared					
	Unweighted			Weighted			Unweighted			Weighted		
	T	C	I	T	C	I	T	C	I	T	C	I
1978	6.91	2.94	3.97	7.00	3.73	3.27	1.00	0.53	0.47	1.04	0.68	0.37
1997	5.23	2.28	2.94	4.68	2.78	1.90	0.52	0.25	0.27	0.36	0.22	0.14
78/97	0.76	0.78	0.74	0.67	0.75	0.58	0.52	0.47	0.58	0.35	0.33	0.38

Deviations from national structure:

	Absolute						Squared					
	Unweighted			Weighted			Unweighted			Weighted		
	T	C	I	T	C	I	T	C	I	T	C	I
1978	7.15	2.95	4.19	6.88	3.42	3.46	1.05	0.51	0.54	0.98	0.56	0.42
1997	5.37	1.96	3.41	4.06	1.99	2.07	0.57	0.23	0.34	0.30	0.14	0.17
78/97	0.75	0.66	0.81	0.59	0.58	0.60	0.55	0.45	0.64	0.31	0.24	0.39

T = total, C = coastal, I = interior.

TABLE IV
VARIANCE DECOMPOSITION—PRIMARY VERSUS SECONDARY INDUSTRY

	Variance of:			Covariance of:	
	$\ln \left(\frac{P_P Q_P}{P_S Q_S} \right)$	$\ln \left(\frac{P_P Q_P / L_P}{P_S Q_S / L_S} \right)$	$\ln \left(\frac{L_P}{L_S} \right)$	$\ln \left(\frac{P_P Q_P / L_P}{P_S Q_S / L_S} \right), \ln \left(\frac{L_P}{L_S} \right)$	
China (28 provinces*)					
1978	0.71	0.12	0.56		0.01
1997	0.53	0.15	0.73		-0.17
United States (50 states & D.C.)					
1977	1.18	0.31	0.63		0.12
1994	0.72	0.32	0.37		0.01
China (14 provinces#)					
1952	1.54	0.44	0.85		0.13
1965	0.98	0.16	0.64		0.09
1978	0.72	0.16	0.63		-0.04
1997	0.58	0.20	0.68		-0.15

*All, but Tianjin and Zhejiang.

#Inner Mongolia, Liaoning, Jilin, Heilongjiang, Shanghai, Jiangxi, Henan, Hunan, Guangdong, Sichuan, Guizhou, Yunnan, Qinghai, and Ningxia.

- **Figure III.** The share of **secondary industry** becomes dramatically compressed across provinces. Excluding outliers, the range falls from about **34%–77%** in 1978 to about **37%–54%** in 1997.

- The primary-sector share also converges, while the service share rises in all provinces and becomes more dispersed because of urban outliers such as Beijing and Shanghai.
- **Figure IV.** Convergence in shares does *not* mean convergence in absolute output per capita. In fact, the standard deviation of primary-sector output per capita **rises by about 37%** between 1978 and 1997, while dispersion in secondary output per capita falls.
- **Main point.** The paper wants readers to distinguish clearly between **convergence in composition** and **convergence in levels**. Trade-barrier stories speak most directly to the former.

5.4 Table III and Figure V: interregional price dispersion

TABLE III
SUMMARY OF PRICE DATA

	Retail prices	Agricultural purchase prices	Industrial materials prices	Agricultural market prices
Geographic unit	city	province	city	city
Frequency	annual	annual	monthly	monthly
# of products	305	130	49	33
# of time periods	8	8	110	71
# prod. $\times$ time combinations*	1774	690	3853	1971
# of price reports*	48717	10361	79824	46135

*Product $\times$ time period combinations and number of city/provincial price observations used in the analysis (products with less than five price observations in a time period are eliminated from the sample).

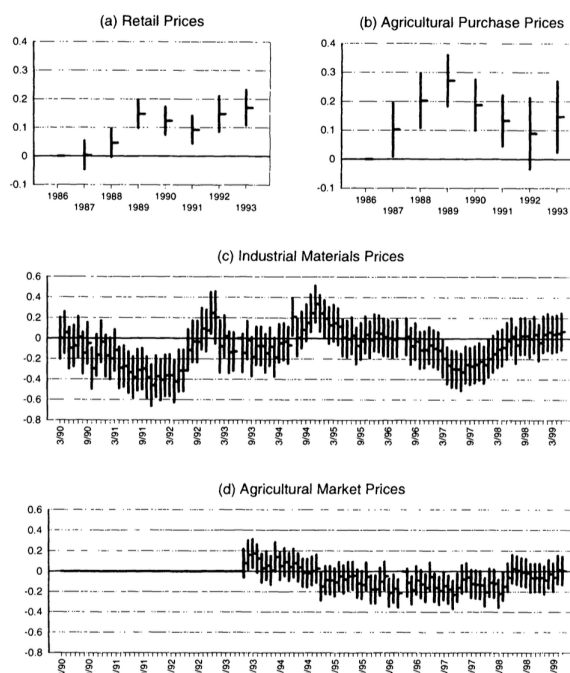


FIGURE V
Trends in the Standard Deviation of ln Product Prices
(time dummies plus or minus two standard errors)

- **Read.** Table III summarizes the four price datasets. The annual retail-price panel has **305 products**; the agricultural purchase-price panel has **130**; the monthly industrial-materials panel has **49**; the monthly agricultural-market-price panel has **33**.
- Figure V estimates time patterns in the cross-regional standard deviation of log prices. Retail price dispersion rises rapidly from 1986 to 1989. Agricultural purchase-price dispersion also rises overall, though less cleanly.
- In the monthly data, dispersion falls and then rises again, with the mean monthly industrial-materials dispersion increasing by about **23%** from 1991 to 1993.
- During the 1990s, price dispersion fluctuates substantially but shows **no clear trend toward re-integration**.
- **Main point.** Prices do not behave as one would expect in a steadily integrating internal market.

TABLE IV
VARIANCE DECOMPOSITION—PRIMARY VERSUS SECONDARY INDUSTRY

	Variance of:			Covariance of:
	$\ln \left(\frac{P_P Q_P}{P_S Q_S} \right)$	$\ln \left(\frac{P_P Q_P / L_P}{P_S Q_S / L_S} \right)$	$\ln \left(\frac{L_P}{L_S} \right)$	$\ln \left(\frac{P_P Q_P / L_P}{P_S Q_S / L_S} \right), \ln \left(\frac{L_P}{L_S} \right)$
China (28 provinces*)				
1978	0.71	0.12	0.56	0.01
1997	0.53	0.15	0.73	-0.17
United States (50 states & D.C.)				
1977	1.18	0.31	0.63	0.12
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1978	0.72	0.16	0.63	-0.04
1997	0.58	0.20	0.68	-0.15

*All, but Tianjin and Zhejiang.
#Inner Mongolia, Liaoning, Jilin, Heilongjiang, Shanghai, Jiangxi, Henan, Hunan, Guangdong, Sichuan, Guizhou, Yunnan, Qinghai, and Ningxia.

5.5 Table IV: variance decomposition and the failure of standard market explanations

- **Read.** For China's 28 provinces, the variance of relative primary-to-secondary output falls from about **0.71** in 1978 to **0.53** in 1997.
- But over the same period, the variance of relative nominal labor productivity rises from about **0.12** to **0.15**, and the variance of relative labor allocation rises more sharply from about **0.56** to **0.73**.
- The covariance term turns substantially more negative, from about **0.01** to **-0.17**. Output convergence therefore occurs only because worsening covariance offsets rising dispersion in fundamentals.
- The U.S. comparison shows a very different pattern: when relative output converges across states, labor allocations also become more similar. The long Chinese series for 14 provinces shows that the reform-era mechanism is unlike the early postwar one.
- **Main point.** The convergence in China's output structure is not being driven by clean convergence in comparative advantage or market integration.

5.6 Figure VI and Table V: agriculture, weather, and perverse factor allocation

- **Figure VI.** In 1978, grain yields and agricultural labor intensity are positively related across provinces. By 1997 that relation essentially disappears. Provinces that were initially most productive in agriculture experience the sharpest relative declines in labor intensity.
- **Table V, yield regression.** Grain yields rise with irrigation and fertilizer and with moderate rainfall, but not with labor intensity or machinery power once other controls are included.
- **Table V, weather interactions.** Provinces with better agricultural weather experience systematically slower growth in irrigation, fertilizer, power, labor, agriculture's labor share, and grain output per capita. The coefficients on the interaction term *Wtime* are all negative and mostly highly significant.

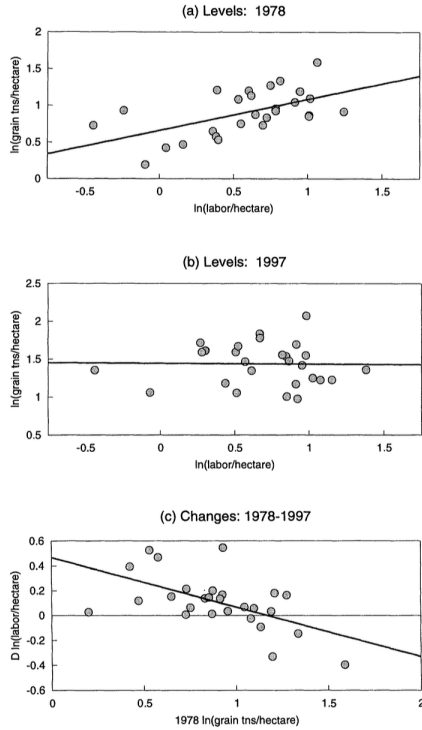


FIGURE VI
Factor Intensity and Productivity (27 provinces)

TABLE V
PRODUCTIVITY, FACTOR ALLOCATIONS, AND THE WEATHER
(ANNUAL OBSERVATIONS, 29 PROVINCES, 1985 TO 1997)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Yield	Irri	Fert	Power	Labor	Lshare	OutputC	Yield
With provincial dummies:								
Year	.009** (3.1)	.019** (8.3)	.091** (33.2)	.049** (20.5)	.006** (3.7)	-.004** (7.6)	.027** (11.4)	.026** (5.9)
Irri		.309** (6.3)						.199** (4.5)
Fert		.214** (6.2)						.155** (4.3)
Power		-.043 (1.0)						-.080* (1.9)
Labor		-.091 (1.5)						-.148* (2.4)
Prec		.021** (2.6)						
Prec2		-.001** (2.7)						
CVPrec		-.040* (1.9)						
Wtime		-.233** (4.8)	-.549** (9.4)	-.194** (3.8)	-.155** (4.4)	-.080** (6.4)	-.426** (8.5)	-.260** (5.6)
R ²	.942	.958	.954	.970	.975	.984	.922	.944
Without provincial dummies:								
Year		.019* (1.8)	.091** (10.7)	.050** (4.0)	.006 (0.7)	-.004 (1.0)	.027** (3.3)	
Weather		-.540 (0.3)	12.4** (9.6)	-3.27* (1.8)	7.34** (5.3)	.790 (1.2)	2.24* (1.8)	
Wtime		-.233 (1.0)	-.549** (3.0)	-.198 (0.8)	-.155 (0.8)	-.080 (0.9)	-.426* (2.5)	
R ²		.027	.522	.136	.171	.029	.031	
N	374	377	377	376	377	377	377	376

Yield = $\ln(\text{grain output}/\text{grain sown area})$; Irri = $\ln(\text{irrigated area}/\text{total sown area})$; Fert = $\ln(\text{chemical fertilizers}/\text{total sown area})$; Power = $\ln(\text{total power of agricultural machinery (kw)}/\text{total sown area})$; Labor = $\ln(\text{employment in primary industry}/\text{total sown area})$; Prec = average monthly rainfall (cm); Prec2 = Prec squared; CVPrec = coefficient of variation of monthly rainfall; Lshare = primary employment/total employment; OutputC = $\ln(\text{grain output per capita})$; Weather = mean value of weather variables multiplied by coefficients in column (1); Wtime = Weather * (year - 1985); *Significant at the 10 percent level; **Significant at the 1 percent level. T-statistics are in parentheses. The sample size increases slightly between column (1) and columns (2)-(8) as, using the mean value of weather, I do not need observations on weather variables in each period.

- The paper's back-of-the-envelope interpretation is economically meaningful: a one-standard-deviation increase in weather quality is associated with roughly **1 percentage point** slower annual growth in fertilizer use and grain output per capita, and roughly **1/2 percentage point** slower annual growth in irrigation, machinery power, and labor.
- **Main point.** The provinces with better agricultural comparative advantage were pulling resources out of agriculture rather than drawing them in.

5.7 Figure VII: the long historical comparison

- **Read.** Figure VII repeats the yield-versus-labor-intensity scatter for fifteen provinces in **1952, 1957, 1965, 1978, 1988, and 1997.**
- The positive relationship between agricultural labor intensity and yields is present throughout the pre-reform period and still visible in 1978.
- By 1988 it is much weaker, and by 1997 it is essentially gone.
- **Main point.** The disappearance of a sensible factor-productivity relationship is not a timeless feature of China. It is a reform-era development.

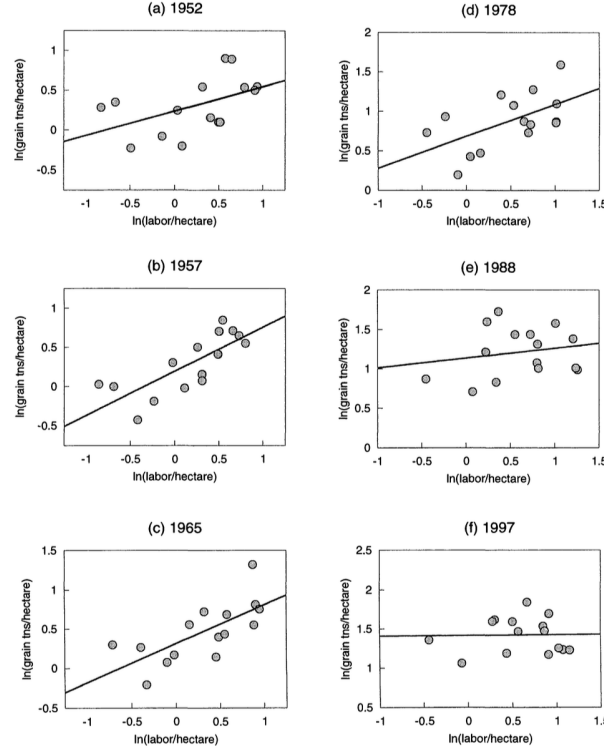


FIGURE VII
Factor Intensity and Productivity (15 provinces)

6 Short summary

- **Method.** The paper combines political-economy narrative with descriptive evidence on output structure, price dispersion, sectoral variance decompositions, and agriculture-weather regressions.
- **Identification.** There is no single instrumental-variable design. Instead, the paper builds a cumulative case by showing that multiple data patterns are jointly inconsistent with standard stories of market integration and comparative advantage.
- **Application.** Reform-era China, especially the interaction between decentralization, fiscal incentives, rural industrialization, and interregional trade barriers.
- **Main result.** Reform did not simply remove old distortions; it generated new ones. China's provinces converged in output structure, diverged in prices and factor allocations, and moved resources away from weather-based agricultural comparative advantage, consistent with growing internal fragmentation.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper argues that China's reform era should not be understood as a clean transition from plan to market.

- Instead, reform devolved power to local governments, which had both the incentives and the means to create new distortions in order to protect local industrial interests.
- The cumulative evidence suggests that China became more open internationally while becoming more fragmented internally.

7.2 Economic intuition

- Partial reform changes who benefits from distortion. Once local governments gain from protecting local industry, they use control over materials, prices, finance, and administration to defend those rents.
- This creates a dynamic in which distortions beget distortions: initial liberalization opens opportunities for rent capture, and those opportunities generate new protectionist behavior.
- The result can be rapid growth together with serious internal inefficiency, because local devolution may be better than central planning overall, even if it does not produce a well-integrated domestic market.

7.3 Takeaway

This is a paper about **how reform can go sideways**. Young's central lesson is that a distorted economy does not necessarily move monotonically toward efficiency once controls are loosened. If prices, fiscal authority, and political control remain partially distorted, decentralization can replace one planner with many local planners, each defending its own protected market. That is the deeper meaning of the paper's title: in a partially reformed economy, the path to efficiency is narrow, and small policy asymmetries can generate large and persistent new distortions.